

AGENTS.md

Guidance for automated coding agents working on `stylometric-transfer`

This document describes the **purpose, architecture, conventions, constraints, and development roadmap** for the `stylometric-transfer` project so that an automated agent can reliably continue development without prior conversation context.

1. Project Mission

`stylometric-transfer` implements:

- **Stylometric profiling** — extracting an explicit, interpretable style model from an author's writing corpus
- **Author-conditioned style transfer** — rewriting text to match that style while preserving meaning

The defining design principle is:

Explicit, interpretable, versionable style models rather than fine-tuning or opaque embeddings.

The system must:

- Remain inspectable by humans
 - Keep the style model as JSON
 - Avoid fine-tuning or hidden representations
 - Preserve meaning strictly in rewriting
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2. Conceptual Terminology (Canonical)

Use the following terms consistently in code, docs, and comments:

- **Stylometry** — quantitative analysis of writing style
- **Stylometric profile / style fingerprint** — the JSON artifact
- **Style transfer** — rewriting while preserving semantics
- **Author-conditioned generation** — generation guided by fingerprint
- **Explicit style model** — rule- and feature-based JSON

Avoid ambiguous terms like "clone" in public documentation.

3. Current Architecture

Files

- `fingerprint_style.py`

- Input: compressed corpus archive (.zip, .tar*)
- Output: style_fingerprint.json
- Responsibilities:
 - Extract archive
 - Read text files
 - Compute stylometric measurements locally
 - Filter out blockquotes, reference sections, footnotes, inline citation markers, and boilerplate notices (copyright/terms/privacy) from style analysis
 - Detect fiction vs non-fiction; in non-fiction, multi-word quotations are excluded from profiling (override with --fiction / --non-fiction)
 - Strip embedded BASE64 images from prompts
 - Select representative excerpts
 - Call LLM to synthesise fingerprint JSON
 - If chunking excerpts, merge partial fingerprints via a dedicated LLM merge prompt
 - Prefilter likely proper-name phrases (honorifics + capitalization-ratio heuristics) and rank phrase candidates with the LLM to drop proper names
 - Optionally rank rare-word candidates with the LLM (shared with common-phrase validation) to de-prioritize proper names
 - Validate common phrases via a separate LLM pass to remove OCR/citation noise (with deterministic prefilters for proper names, entity blacklist matches, and date patterns; disable with --no-phrase-validation)
 - Repair invalid JSON if necessary
 - Normalize verbose/duplicative controls.rewrite_policy clauses and filter priority_order to short tokens before writing the fingerprint
 - Normalize lexicon spelling to a US baseline (except literal hard avoids)
- CLI short flags: -c (config, optional; defaults to ./config.llm.json if present, else next to script), -a (archive), -o (out), -v (verbose)
- Extra: --max-prompt-tokens overrides chunking threshold
- Defaults: if --profile-id or --author-name are omitted, both default to the output filename without the .json extension

- apply_fingerprint.py

- Input: fingerprint JSON + Markdown file
- Output: rewritten Markdown + deviations report
- Responsibilities:
 - Measure input text
 - Detect fiction vs non-fiction; in non-fiction, multi-word quotations are preserved verbatim (override with --fiction / --non-fiction)
 - Preserve blockquotes, reference sections, footnotes, and inline citations verbatim (excluded from style transfer)
 - Strip embedded BASE64 images before prompt and re-insert after rewrite
 - Call LLM with fingerprint + measurements
 - Enforce preservation of meaning

- Score style compliance locally and retry once by default with delta feedback (disable with `--no-style-retry`)
 - Apply `general-guidelines.md` humanizer rules when available, using an LLM parser by default then deterministically filtering out conflicts (disable with `--no-humanizer-llm-parse` or `--no-humanizer-guidelines`)
 - Cache parsed humanizer rules in `humanizer_rules.cache.json` (script directory) and re-parse only when guidelines change
 - Normalize verbose/duplicative `controls.rewrite_policy` clauses and filter `priority_order` when loading a fingerprint
 - Normalize lexical avoidance checks to US spelling for matching, then apply local spelling to final output
 - Return rewritten text and deviations
 - CLI short flags: `-c` (config, optional; defaults to `./config.llm.json` if present, else next to script), `-f` (fingerprint; adds `.json` if missing), `-i` (input), `-o` (out), `-v` (verbose)
 - Extra: `--max-prompt-tokens` overrides chunking threshold
 - Overrides: `--1st-person` / `--2nd-person` / `--3rd-person` force narrative voice regardless of fingerprint
- `scripts/fingerprint_style.sh` and `scripts/apply_fingerprint.sh`
 - Bash wrappers around the Python entry points
 - Pass all CLI args through unchanged
 - `show_fingerprint.py`
 - Generates a standalone HTML dashboard for a fingerprint JSON
 - `prompts.json`
 - Externalized prompt templates used by both Python entry points
 - Located next to the Python scripts; loaded at runtime
 - `config.llm.json`
 - Stores API configuration
 - OpenAI-compatible
 - `max_prompt_tokens` controls chunking for large prompts (defaults to `max_tokens`)
 - `max_retries` / `backoff_*` control exponential backoff on transient LLM failures
 - Optional `lexicon_hints.json` can be used to inject preferred/avoided phrases into fingerprinting
 - `config.tunables.json`
 - Overrides humanizer conflict thresholds for `apply_fingerprint.py`
 - See README for per-field explanations and defaults
 - Includes mandatory humanizer controls (em-dash ban, emoji policy)
 - Includes bounded stochastic variance controls (seeded micro-operations)

- Includes style retry controls (delta-feedback retries)
 - Includes humanization metric weights for the aggregate 0–100 score
 - Includes corpus-derived humanization baseline settings (rolling windows embedded in fingerprints for auditability; stripped from what the LLM sees during rewriting)
 - Includes humanization controller overlays (per-chunk target variation derived from baseline)
 - Includes chunking caps (max input tokens per chunk), split strategy (`chunk_split_on`), and rolling chunk summaries for semantic continuity
 - Includes minimum chunk counts when perturbations are enabled
 - Includes chunk-recovery split controls when the LLM repeatedly returns invalid output
 - Includes variance-aware chunk sizing (smaller chunks for higher-variance styles)
 - Includes lexical signal limits (rare word list size)
 - Includes lexical avoidance limits (rare word list size)
 - Includes controls normalization (rewrite-policy de-dup + priority-order filtering)
 - Includes fiction-detection thresholds (quote span/ratio heuristics)
 - Includes section-restore controls (fuzzy heading match + restoration caps)
 - Includes optional sanity checks such as line/word/paragraph count change warnings
 - The current tunables may be embedded into fingerprints as `metadata.extraction.tunables_snapshot` for auditability
- `config.avoid.txt`
 - Optional global avoid list (one word/phrase per line)
 - Applied during fingerprinting (via lexicon hints) and merged into `lexicon.avoid_words` (hard avoids) during application
 - Entries are literal (no spelling normalization); include any local spelling variants you need
 - `config.common_words.txt`
 - Optional common-word list (one word per line)
 - Used to derive `measurements.lexical_avoidance.rare_words` (common words absent from the corpus)
 - Supports `word <zipf_frequency>` format to prioritize higher-frequency words
 - These omissions are intended as soft avoids (`lexicon.avoid_words_soft`) rather than hard bans
 - `config.entity_blacklist.txt`
 - Optional entity blacklist (people, places, organizations; one per line)
 - Used to suppress proper-name phrases during common-phrase validation

Data Flow

Corpus → Voice-filtered local stats → LLM synthesis → Fingerprint JSON
 Fingerprint + Draft → Voice-filtered local stats → LLM rewrite → Styled Markdown

4. Design Constraints (Critical)

These must always be preserved.

4.1 Explicit Models Only

- The style model MUST remain:
 - JSON
 - Human readable
 - Editable
 - Versionable

Do NOT introduce:

- Fine-tuning
- Embedding-only representations
- Hidden latent style vectors without interpretation

4.2 Meaning Preservation

In rewriting:

- No new facts
- No new claims
- No new examples
- No semantic drift

Deviation reporting is mandatory when conflicts occur.

4.3 Interpretable Metrics

Local measurements must:

- Remain simple and explainable
- Use distributions over single averages where possible
- Prefer sentence/paragraph/lexical statistics

Avoid black-box scoring models unless clearly labeled optional.

5. Style Fingerprint Schema (High-Level)

The fingerprint JSON contains these top-level keys:

- `schema_version`
- `profile_id`
- `metadata`

- **measurements** (verbatim local statistics)
- **targets** (style constraints)
- **lexicon** (preferred / avoided words & phrases)
- **templates** (syntactic + rhetorical moves)
- **controls** (priority + strictness + rewrite policy)
- **validators** (scoring weights + checks)
- **derived_instructions** (compiled prompts)

The schema is **extensible**, but backward compatibility should be preserved when possible.

Note: **metadata.corpus** includes **document_count** and **documents** (per-document metadata list, including title when available).

6. Prompting Strategy (Canonical)

6.1 Fingerprint Construction

Process:

1. Filter out non-author voice content (blockquotes, references/footnotes, inline citations)
2. Compute local measurements
3. Validate common phrases with an LLM pass to remove OCR/citation noise (disable with **--no-phrase-validation**)
4. Select representative excerpts
5. Include optional **lexicon_hints.json** if present
6. Send both to LLM with:
 - Schema hint
 - JSON-only requirement
 - Controlled vocabulary instructions

LLM must:

- Embed measurements verbatim
- Infer stylistic targets from measurements + excerpts
- Produce derived instructions for reuse

6.2 Application / Rewriting

Prompt must:

- Include full fingerprint JSON
- Include local measurements of input (computed on author-voice text only)
- Enforce:
 - Preservation of meaning
 - Markdown validity
 - Priority order in constraints

LLM output format is JSON:

```
{
  "final_markdown": "...",
  "deviations": [...],
  "self_check": {...}
}
```

7. Coding Conventions

7.1 API Client

- Use OpenAI-compatible REST calls
- Prefer `/v1/chat/completions` unless migrated
- Centralize retry/backoff logic

Recommended improvements (future work):

- Exponential backoff for 429 / 5xx
- Streaming optional flag

7.2 Error Handling

Always handle:

- Invalid JSON output
- Partial JSON
- Code-fenced JSON

Current strategy:

- Attempt strict parse
- If failure, call LLM repair mode

Preserve this pattern.

8. Measurement Layer (Do Not Over-Engineer)

Current measurements include:

- Sentence length histogram
- Paragraph length histogram
- One-sentence paragraph rate
- Punctuation rates per 1000 words
- Contraction rate

- US vs Canadian spelling heuristic (English-only)
- Dash / ellipsis counts
- Frequent bigrams / trigrams
- Function-word profile
- Rare-word signals (words the author rarely uses)
- Lexical avoidance list derived from a built-in common-words set (common words the author rarely uses)
- Stance signals (hedging/boosting/pronouns)
- Sentence-opener and transition templates
- Rhetorical move signals (claim/evidence/counterpoint/concession/synthesis)
- Paragraph cadence (opening/closing sentence length stats)
- Epistemic profile (speculative/probabilistic/assertive/directive)
- Syntax texture (subordinate/parenthetical/appositive rates)
- Discourse marker position (start vs mid-sentence rates)
- Repetition signals (bigram/trigram repeat rates)

Schema note:

- `measurements.orthography_signals.spelling_variant` records the spelling heuristic output
- `targets.persona.pronoun_preferences` records preferred pronoun sets (if provided)

Guidelines:

- Prefer approximate signals over fragile exact metrics
- Avoid heavy NLP pipelines unless optional
- Do NOT require spaCy / transformers by default
- Exclude non-author voice regions (blockquotes, references/footnotes, inline citations) from measurements and excerpts

Future additions allowed:

- POS tag distributions (optional)
- Readability indices
- Function word profiles

9. Ethics & Safeguards

Always preserve:

- Intended use: personal writing, self-modeling, editing assistance
- Avoid impersonation of living authors

Controls in fingerprint:

- `do_not_imitate_living_author = true`
- `copyright_sensitive = true`

Agents must NOT add features that:

- Automate impersonation
 - Mask authorship
 - Remove deviation reporting
-

10. Roadmap (Agent Guidance)

High-value next steps, in priority order:

Phase 1 — Reliability

- ☐ Add retry + exponential backoff wrapper for API calls
- ☐ Add request timeout + cancellation handling
- ☐ Add logging verbosity flag

Phase 2 — Validation

- ☐ Add JSON Schema file (`schema/style_fingerprint.schema.json`)
- ☐ Validate fingerprint outputs with `jsonschema`
- ☐ Emit warnings on missing or null critical fields

Phase 3 — Scoring & Feedback

- ☐ Add post-rewrite scoring against fingerprint
- ☐ Compute divergence metrics (sentence length, punctuation)
- ☐ Emit compliance score (0–1)

Phase 4 — Tooling

- ☐ Batch rewrite mode
- ☐ Batch fingerprint mode
- ☐ Diff visualizer (original vs styled)
- ☐ CLI subcommands (`stylometric fingerprint`, `stylometric apply`)

Phase 5 — Generation

- ☐ Add `generate_from_fingerprint.py`
- ☐ Support outlines / prompts conditioned on fingerprint

Refactoring (Future)

- ☐ Consolidate additional shared helpers (masking/placeholder, regex utilities, LLM client wrappers) into a shared module alongside `utils.py` to reduce duplication.
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11. Testing Strategy

Current state: lightweight regression tests + smoke test

Recommended additions:

- Golden corpus test:
 - Run fingerprint on fixed small corpus
 - Verify stable JSON fields
- Rewrite invariants:
 - Word count within tolerance
 - Named entities preserved
 - No added sentences unless allowed
- JSON validity tests

Current smoke test:

- `tests/run_smoke.sh` — regression suites by default (no LLM calls); end-to-end pipeline + LLM connectivity check when `--llm-tests` is provided (requires valid `config.llm.json`)

Regression suite:

- `tests/test_v1_1_0_regression.py` (run via `./tests/run_v1_1_0_regression.sh`; executed automatically by `run_smoke.sh`)
- `tests/test_v1_5_X_regression.py` (run via `./tests/run_v1_5_X_regression.sh`; executed automatically by `run_smoke.sh`)
- `tests/test_v1_7_X_regression.py` (run via `./tests/run_v1_7_X_regression.sh`; executed automatically by `run_smoke.sh`)

12. Naming & Public Interface

Preferred public names:

- Project: `stylometric-transfer`
- Artifact: **Style Fingerprint** or **Stylometric Profile**
- Process: **Stylometric profiling** + **Author style transfer**

Avoid marketing terms like:

- “Magic”
- “Clone”
- “Deep copy”

13. Summary for Agents

If you are an automated agent continuing this project:

You are building an **interpretable stylometric profiling and style-transfer system** with these invariants:

- Explicit JSON style models
- Local statistical grounding
- LLM used only for synthesis and rewriting
- Meaning preservation is sacred
- Deviations must be reported
- Human inspectability is a core feature

Any new feature should improve:

- Reliability
- Interpretability
- Reproducibility
- Control

—not raw imitation fidelity alone.

AGENTS.md — operational context for stylometric-transfer